

第三章补充题参考答案

7. (补充题 1) 证明: 若 $\hat{F}$ 和 $\hat{G}$ 是 Hermitian 算符而且满足 $\hat{F}\hat{G}=\hat{G}\hat{F}$, 那么 $\hat{F}\hat{G}$ 也是 Hermitian 算符.

【证明】 $\hat{F}$ 和 $\hat{G}$ 是 Hermitian 算符, 有

$$\begin{aligned}\hat{F}^\dagger &= \hat{F}, \quad \hat{G}^\dagger = \hat{G} \\ (\hat{F}\hat{G})^\dagger &= \hat{G}^\dagger \hat{F}^\dagger = \hat{G}\hat{F}\end{aligned}$$

由已经 $\hat{F}\hat{G}=\hat{G}\hat{F}$, 得

$$(\hat{F}\hat{G})^\dagger = \hat{F}\hat{G}.$$

8. (补充题 2) 定义 $L_+ = L_x + iL_y$, $L_- = L_x - iL_y$, 则有 $[L_z, L_\pm] = \pm\hbar L_\pm$ (其中 L_x 、 L_y 和 L_z 是轨道角动量分量算符).

【证明】

$$\begin{aligned}[\hat{l}_z, \hat{l}_\pm] &= [\hat{l}_z, \hat{l}_\pm] = [\hat{l}_z, \hat{l}_x \pm i\hat{l}_y] = [\hat{l}_z, \hat{l}_x] \pm i[\hat{l}_z, \hat{l}_y] \\ &= i\hbar\hat{l}_y \pm i(-i\hbar\hat{l}_x) = i\hbar\hat{l}_y \pm \hbar\hat{l}_x = \pm\hbar(\hat{l}_x \pm i\hat{l}_y) = \pm\hbar\hat{l}_\pm\end{aligned}$$

9. (补充题 3) 在 L_z 的本征态中, 求解 $\bar{l}_x$ 和 $\bar{l}_y$.

【解答】 设 $|\phi\rangle$ 为 $\hat{l}_z$ 的本征态, 则有 $\hat{l}_z|\phi\rangle = l_z|\phi\rangle$, 其中 l_z 为本征值.

根据对易关系 $\hat{l}_x = \frac{1}{i\hbar}[\hat{l}_y, \hat{l}_z]$, 有,

$$\begin{aligned}\bar{l}_x &= \frac{1}{i\hbar}\langle\phi|[\hat{l}_y, \hat{l}_z]|\phi\rangle = \frac{1}{i\hbar}\langle\phi|\hat{l}_y\hat{l}_z|\phi\rangle - \frac{1}{i\hbar}\langle\phi|\hat{l}_z\hat{l}_y|\phi\rangle \\ &= \frac{1}{i\hbar}l_z\langle\phi|\hat{l}_y|\phi\rangle - \frac{1}{i\hbar}l_z\langle\phi|\hat{l}_y|\phi\rangle = 0.\end{aligned}$$

同理

$$\bar{l}_y = \frac{1}{i\hbar}\langle\phi|[\hat{l}_z, \hat{l}_x]|\phi\rangle = 0.$$